

Nested oscillations in superdiffusive dynamics of visual searching

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Declaration

I, *Jude Luca Metcalf*, declare that this thesis is submitted in partial fulfilment of the requirements for the conferral of the degree *Bachelor of Science (Honours)*, from the University of Sydney, is wholly my own work unless otherwise referenced or acknowledged. This document has not been submitted for qualifications at any other academic institution.

Jude Luca Metcalf

August 19, 2026

Abstract

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Introduction

The brain is among the most widely studied subjects of research, and yet many neural behaviours remain elusive to scientists. Vision is an important neural process, and provides one of the first gates of attention to the human brain as the visual system actively interacts with the environment. A large part of the brain is devoted to vision, the visual cortex containing between 4-6 billion of the 30 billion neurons of the neocortex, up to 5 times the size of other primates [1, 2]. In particular, search tasks reveal complex visual dynamics and invite research into mechanisms of attention and brain function through a functional understanding of gaze patterns. To study these patterns of gaze dynamics, methods adapted from statistical physics, network science, and non-equilibrium thermodynamics have been found to explain and provide understanding of neural computation. Recent approaches to modelling human gaze during visual search have predominantly relied on training large neural networks to predict gaze trajectories [3–8] and saliency maps from an image for human observers [9–14]. Such study has fostered developments in the broader field of artificial intelligence, where inspiration from the behaviour of the human brain is often explicit. Most notably, the DeepGaze III project developed by Kümmerer, Bethge, and Wallis [15] has yielded high accuracy in predicting the location of the fixations of human vision given the image viewed, the fixation history, and biased patterns of searching [9, 15, 16]. Such trained deep learning models of eye movements have allowed researchers to test predictions concerning the behaviour of visual searching, such as how simple low-level contrast is often enough to predict fixations [17], or how late-time fixations quickly become dominated by gaze bias and decouple from the geometry of the image viewed [18]. Such techniques have allowed us to understand much from the large amount of empirical data gathered from search tasks.

However, despite the predictive success of these models, a mechanistic, dynamical account of visual searching beyond fixation prediction remains largely unexplored. Such a unifying formulation should account for a large range of well-documented patterns of visual search behaviour. For instance, search dynamics are identified by the spatiotemporal patterns of the excursions of visual focus called saccades, the dynamical behaviour during fixation between such attentional saccades, the preferential searching after interesting areas of high saliency, the inhibition of attention in previously explored areas, and the emergence of timescales of attention. To be valuable, such a formulation should be biologically plausible, and yield testable predictions and functional explanations of novel or unexplained aspects of visual dynamics.

Chapter 1: Survey of the Literature

This thesis aims to further our understanding of the details of biological neural computation through understanding the dynamics of visual searching. There is growing success in turning methods from statistical physics towards the area of computational neuroscience, and major contributions to this project are described here. In particular, the rich scale-free activity of neural circuits inspires similar analysis to the dynamics of gaze trajectories, alongside the information-theoretic pressure of the visual system to process data near-optimally. We aim to understand these behaviours as an efficient method of gathering information. In Section 1.1 we introduce the framework of stochastic modelling, Lévy motion, and the associated anomalous superdiffusion which informs the dynamics of the FNS sampler. Section 1.2 then provides the link between stochastic dynamics and computation sampling, both in biological and abstract contexts. Section 1.3 provides the background for much of the behavioural data we present and model in this thesis, and the previous methods used to understand and model the complex dynamics found in gaze trajectories especially.

1.1 Statistical physics

1.1.1 Brownian motion

Statistical modelling has been fundamental in understanding complex systems: separating the dynamics of macroscopic observables into deterministic terms and stochastic terms reduces high-dimensional systems to a small number of variables of interest [19, 20]. Across physics, economics, geology, climate science, machine learning, and neuroscience, stochastic processes are often the most natural, tractable, and accurate modelling choice when a system is too complex to model in full. The universality of these stochastic models is explained by the application of the central limit theorem, where the sum of very many small interactions over macroscopic time limits to a Gaussian random variable. This techniques turns out to be generalisable to a wide class of coupled nonstationary processes out of equilibrium, as described in Izvekov [21], whose techniques can describe the mean effect of a neuronal spiking patterns as demonstrated in La Camera [22]. Such processes thus produce the most ubiquitous stochastic process: the Wiener process W_t, equivalently called Brownian motion, which has stationary and independent Gaussian increments across all non-overlapping time intervals. The Wiener process is a very simple and analytically elegant model of the coarse-grained effect of very many small finite-variance interacting influences. A Wiener process is defined only by its timescale, with a variance over trajectories after time t of t itself. Wiener processes yield time series which are almost surely continuous but nowhere differentiable. The Wiener process is understood as a model of the long-time dynamics of a variable effected by very many small uncorrelated finite variance influences [19]. Classical examples of processes modelled by the Wiener process are: the movement of a macroscopic particle in a non-turbulent fluid [19], the gravitational influence of an orbiting x collection of stars on supermassive black holes [23, 24], and (arguably) the movement of low-volatility stock prices [20].

To include random variables such as Brownian motion in dynamical systems, physicists and mathematicians have developed the tools of stochastic calculus. Brownian motion about a potential landscape $V(\mathbf{x})$ is often described in dynamical systems contexts in terms of the standard first order Langevin equation:

$$d\mathbf{x}_t = -\gamma \nabla V(\mathbf{x}_t)dt + \sqrt{2\gamma}dW_t \tag{1.1}$$

Where the constant γ describes the noise strength of the Brownian motion. To describe how the probability distribution of this walker $p(\mathbf{x}, t)$ evolves through time, the Fokker-Planck equation is used. In the case of Equation (1.1), the Fokker-Planck equation is:

$$\partial_t p(\mathbf{x}, t) = \frac{\gamma}{2}\nabla \cdot (p(\mathbf{x}, t)\nabla V(\mathbf{x}) + \nabla p(\mathbf{x}, t)) \tag{1.2}$$

whose unique stationary distribution is the Boltzmann-Gibbs distribution with $k_B T = 1$:

$$\pi(\mathbf{x}) \propto \exp[-V(\mathbf{x})] \tag{1.3}$$

which yields the important relationship, where a potential can be constructed from a normalised target density via $V(\mathbf{x}) = -\log \pi(\mathbf{x})$. Importantly, the dynamical system defined by Equation (1.1) is ergodic, and thus under Equation (1.2), any initial distribution will converge to $\pi(\mathbf{x})$. The Fokker-Planck equation for the standard Brownian-Langevin system is a diffusion PDE equivalent to the heat equation with a gradient term, a correspondance well-understood before the development of stochastic calculus. The diffusion arising from such small uncorrelated perturbations of a dynamical system leads to the statistical properties which first drove the study of statistical physics, such as normal diffusion and the universality of Brownian motion, which we follow in Section 1.1.2 and Section 1.1.3 respectively, however the neural system described in this thesis diverges strongly from the Brownian paradigm, requiring an understanding of fractional processes.

Further stochastic models

Only recently have the universality of stochastic processes asides from the Wiener process been well-studied. Of interest to this thesis is the case where microscopic perturbations become correlated and yield heavy-tailed statistics. Correlated steps, long-time memory effects, and chaotic microscopic dynamics can result in a stochastic process which does not limit to a Brownian motion [25, 26]. Of interest to us is the fractional *Lévy flight*, a process driven by scale-free noise which span across many orders of magnitude. For instance, a spiking neural network at with diverging activity correlation length, where the centre of mass of neural firing is then subject to large jump discontinuities [22, 27]. Such large jumps in time series' are modelled with to Lévy stable noise, where the increments are drawn independently from a heavy-tailed distribution. This section describes the analytics, stability, and behaviour of such Lévy processes for our current study, and how their diffusive properties differ from the behaviour of Brownian motion.

1.1.2 Anomalous diffusion

A key motivation for the development of such stochastic modelling of phenomena in nature was the study of diffusion. In 1885, the German physiologist Adolf Fick was the first to describe the universality of *normal diffusion* in systems which later would be called Brownian motion, specifically the movement of particles in a fluid [28]. Fick discovered prior to the discovery of the Fokker-Planck equation that they evolve via the heat equation

$$\partial_t p(x,t) = D\nabla^2 p(x,t) \tag{1.4}$$

where D is a scaling constant. Fick shows that particles evolving through this system diffuse with a mean squared displacement (MSD) of $\langle x(t)^2 \rangle = 2Dt$. This MSD linear in time defines normal diffusion. Such normal diffusion is universal to any realisation of Brownian motion. To study such diffusive random walks and their long time behaviour, German physicist Albert Einstein and the British statistician Karl Pearson independently developed the continuous time random walk (CTRW) where a walker draw subsequent step sizes and durations from finite variance distributions. Such models will reproduce this normal diffusive property, and as a result of the central limit theorem, approach Brownian motion. Increasingly, physical systems of interest have been found which diffuse nonlinearly in time, a situation known as anomalous diffusion, with an MSD scaling as:

$$\langle x(t)^2 \rangle \propto t^\beta, \quad \beta \neq 1 \tag{1.5}$$

Such anomalous diffusion yields two cases, subdiffusion $0 < \beta < 1$ where diffusion scales slower than normal, and superdiffusion $1 < \beta < 2$ where diffusion scales faster than normal. The case of $\beta = 2$ is known as ballistic diffusion, as it is found in the deterministic case of constant velocity $x = vt$. Anomalous subdiffusion has been shown to result from heavy-tailed wait-times between

interactions [28], such as in trapped fluid flow [29], controlled diffusion through cell membranes [28], and protein mixing [30].

Anomalous superdiffusion

More importantly to this thesis, superdiffusion has been studied in the contexts of complex exploratory biological systems. Superdiffusion has a strong link to the problem of a blind search for scarce resources, however the true computational benefit of superdiffusion has become a subject of debate in the literature. On one side, the Lévy foraging hypothesis posits that the standard superdiffusive CTRW, the Lévy walk, is an optimal strategy when searching for sparse and randomly distributed resources. This result was first reported in a letter to the journal Nature in 1999 by Viswanathan et al. [31]. However, recent physicists and biologists have questioned if this is instead an artefact of improper modelling, an argument led by described most strongly this year by Pyke, Kepert, and Chivers [32]. In the original paper proposing the Lévy foraging hypothesis a Lévy walk with superdiffusion $\langle x(t)^2 \rangle \propto \frac{t^2}{\ln(t)}$ on the transition to ballistic diffusion was found to be optimal in a search for sparse resources [31]. Further, the optimality of the superdiffusive Lévy walk has been reproduced in diverse functional search models [33–35], in a population of diverse searching agents with a range of superdiffusive exponents [36], and have been reported in the empirical search paths of biological organisms from the flights of drosophila flies [35, 37], albatross' searches [38, 39], and strongly across many marine predators [40, 41]. Despite these analyses, advantages of Lévy walks have been shown to be model-dependent, with the optimal performance of Lévy walks requiring biologically inaccurate depictions of search paths [32, 42, 43]. These arguments marks a split in the treatment of Lévy walks, with groups studying them as informative baseline functional models of abstract search [31, 33, 36, 42], and others led by Graham Pyke presenting this as a misrepresentation of true intelligent organisms [32, 43, 44]. This thesis provides a biologically realistic and well-motivated superdiffusive search for research into a case for optimal Lévy motion. See Reynolds [45] for an in-depth discussion of a richer view of biological Lévy walk superdiffusion.

1.1.3 Lévy stable motion

To mathematically capture the intermittent, scale-free nature of visual search, we turn to the *Lévy flight*: a superdiffusive stochastic process that permits close analysis, due to it's very foundational mathematical definition. Here, the Lévy flight L_t^α is a stochastic process defined through it's increments being drawn from a heavy tailed symmetric α-stable distribution (a $\mathcal{S}\alpha\mathcal{S}$ distribution), where α is called the characteristic exponent and relates to the exponent of the distributions powerlaw tails. $\mathcal{S}\alpha\mathcal{S}$ distributions describe the limit of summation of i.i.d infinite-variance variables, as a result of the generalised central limit theorem, making them ideal to describe a universality class of superdiffusive processes. The α value defining a particular $\mathcal{S}\alpha\mathcal{S}$ distribution is a result of the asymptotic power-law tail of the density function, with the probability density function following $p(x) \propto |x|^{-1-\alpha}$. As a result, the Lévy flight L_t^α after time t is defined by its fractional scaling properties and alpha-stable statistics

$$L_t \overset{d}{=} t^{1/\alpha} L_1, \quad L_1 \sim \mathcal{S}\alpha\mathcal{S} \tag{1.6}$$

where $\overset{d}{=}$ denotes equal in distribution. In the limit $\alpha \to 2$, the Lévy flight reduces to a scaled Brownian Motion $L_t^{\alpha \to 2} = \frac{1}{\sqrt{2}} W_t$. If $1 < \alpha < 2$ the variance of L_t^α diverges and has mean 0, whereas if $0 < \alpha \leq 1$, then both the variance and mean diverge. In analysing the distribution of a Lévy flight, one must generalise the standard Langevin Fokker-Planck equation (1.2) to the Lévy motion using fractional calculus to capture the nonlocal transport of the particle. The details of such an analysis are beyond this thesis, however the fractional Fokker-Planck equation is the tool used to understand the superdiffusion, nonlocality, and scale-free behaviour of trajectories of Lévy flights (see the appendices to [27]). Many of the simple analytic properties of the Wiener process are no longer true of a Lévy flight. Notably, since these flights have infinite variance

in steps sizes, the mean squared displacement diverges, and thus their superdiffusion is better measured using mean fractional displacements

$$\langle x(t)^q \rangle \propto t^{q\beta/2}, \quad q > 0, \quad 0 < \beta \leq 2 \tag{1.7}$$

where q is the order of the mean fractional displacement, and these metrics will converge with $\beta = 1/\alpha$ when $q < \alpha$. As such, for Lévy flights with exponents $\alpha > 1$ superdiffusion is well-characterised by the mean absolute displacement, where $q = 1$ in Equation (1.7).

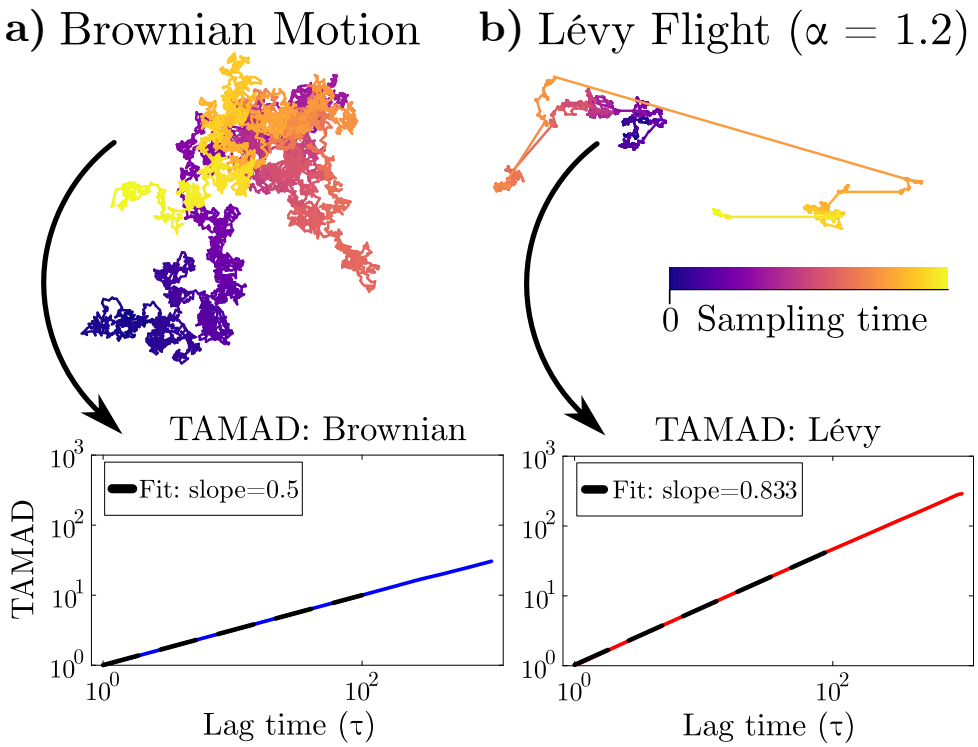


Figure 1.1: Schematic representations of stochastic motion and their respective diffusion. a) Schematic representation of a 2D Brownian motion, along with its time-averaged mean absolute displacement (TAMAD), which is the $q = 1$ mean fractional displacement as in Equation (1.7). Normal diffusion is here present as a power law slope of exponent 1/2, giving diffusive exponent $\beta = 1$ as defined in Equation (1.7). **b)** Schematic of a superdiffusive 2D Lévy flight with characteristic exponent of $\alpha = 1.2$, and a MAD slope of 0.833 corresponding to the theoretical superdiffusive coefficient $\beta = 2/\alpha = 5/3$.

Importantly, trajectories of a Lévy flight are almost surely no longer continuous. Instead, such trajectories are called *càdlàg* (continue à droite, limite à gauche) with respect to time, which describes functions which are right continuous with well-defined left limits [46]. Thus, we say that since all such trajectories produced from a Lévy flight are càdlàg, the Lévy flight itself is called a càdlàg process. Càdlàg processes have been widely studied in the context of mathematics, economics, and physics, and their properties are well understood [47, 48]. It has been shown that Lévy flights are semimartingales, and are well interpreted in integration [46]. Importantly, it follows from the Lévy-Itô decomposition theorem that a Lévy flight can be decomposed as the sum of two processes [49]:

1. A compound Poisson process of large jumps with $|\Delta x| > 1$. Such jumps are exponentially distributed in time, and there are almost surely a finite number of such jumps in any finite time period.

2. A square-integrable martingale (mean 0) jump process with $|\Delta x| < 1$. This process is then finite-variance, and as such is subject to the central limit theorem, meaning that under temporal coarse graining it approaches a Wiener process.

This division is useful in analysing the process in the context of nonlocal stochastic exploration in Lévy flights, which is modelled by the compound Poisson process of large jumps $|\Delta x| > 1$, and the local exploitation dynamics modelled by the square-integrable martingale of small jumps $|\Delta x| < 1$. In the context of visual searching, such a division naturally inspires the division between large jumps of attention, and periods of prolonged attention. As such, we are interested in why such superdiffusive scale-free dynamics may have a powerful in computational advantage in exploring an abstract unknown landscape.

1.2 Abstract computation

The stochastic dynamical exploration of an information landscape is a powerful paradigm of general computation, and is known as probabilistic computation. Such a landscape in this context is a representation of a target probability distribution. Sampling processes represent and evaluate these distributions through the efficient production of informative stochastic samples. Uncertain, high-dimensional, and ambiguous inference problems require efficient and accurate characterisations of target distributions. Sampling methods aim to draw sampled independently from a target distribution. This paradigm has become widely utilised in machine learning and statistics, where generative models learn patterns from sparse samples of large multivariate distributions inferred from observational data [50, 51]. This framework has recently gained significant traction as a basic scheme to describe biological computation. The variability in firing patterns of neural populations would thus maintain an internal model of perceptual uncertainty and evolving understanding, giving a functional role to chaotic neural activity [52–54].

1.2.1 Neural sampling

The framework of neural sampling posits this paradigm that the variability of cortical neural responses is an important functional feature of biological computation. Otherwise regarded as noise, stochastic neural activity patterns are thus presented as samples drawn from a posterior distribution over possible sensory interpretations [55, 56]. The neural sampling hypothesis has been shown to perform and predict network-level empirical patterns of neural activity in the V1, and recently by Terada and Toyoizumi in biologically realistic cortical networks, better than alternative theories [55–58]. Perceptual uncertainty is then encoded by the variability of these cortical responses across time and across multiple trials, with higher uncertainty reflected in wider response distributions [50, 55, 56]. Neural sampling thus provides an interpretation for the way the brain navigates uncertainty at the fundamental scale of neural computation.

Studies into neural sampling show that variability is not strictly a result of explicit stochastic noise, as traditionally believed. Instead, recent work demonstrates that strongly recurrent, balanced inhibitory-excitatory deterministic neural networks can generate the high-dimensional variability which stochastic models are built to capture [56, 58]. Terada and Toyoizumi link this sampling of a posterior distribution to Bayesian learning through training a recurrent network to sample static features and dynamic trajectories, generalising learned priors to novel, uncertain environments through biologically plausible learning [58]. Further biologically plausible implementations of such Bayesian sampling are energy-based models (EBMs) which suggest that fast-spiking interneurons can estimate the local partition function to enable fast, localised sampling dynamics [59]. The sampling hypothesis has thus given a convincing theory of biological neural computation, inspired by the Bayesian brain framework (as described above in **??**), with testable and well-supported evidence that neural variability has an important functional role in computation [55, 57]. Traditional Gaussian and Poisson noise models representing these stochastic network dynamics inherently correspond to continuous struggle to capture the full dynamics of complex interconnected inhibitory-excitatory (IE) biological neuronal network [60–

62]. Further study into the dynamics of these IE networks has thus motivated further functional models of neural activity.

Visual search

This thesis presents visual search tasks as such a paradigm where a sampling model is well motivated, and driven by the underlying neural sampling process as well. The visual system is evolved to actively and efficiently process noisy information and evidence from the external world, and plan future search paths to gather further noisy evidence [14, 63–65]. In the context of visual searching, visually interesting targets with *saliency* as the equivalent of the target landscape, and where the information is located in a target image. Here, saliency is equivalent to the Boltzmann log-likelihood potential from the probability distribution of fixations $p(\mathbf{x})$

$$S \propto \log(p(\mathbf{x})) \tag{1.8}$$

where visual search paths are most likely to be drawn to is a common technique whic [66]. As such, undirected visual searching of an image will show the areas most salient to the brain, allowing insight into the computational sampling mechanism of the visual system. In particular, search tasks require sophisticated integration of the cost and expected information gain of decisions, which invites fruitful decision analysis [67]. Empirical experiments show that eye movements show good agreement with greedy local information maximising fixations, given well-motivated visual sensitivity based on human physiology [64, 65, 68].

Fractional neural sampling

In reality, neural activity has been found to exhibit structure across multiple spatial and temporal scales, often described by scale-free statistics [69–77]. This observation, together with the neural sampling hypothesis, motivates the design and theory of Fractional Neural Sampling (FNS) [27]. FNS describes the complex spatiotemporal dynamics of spiking neural circuits, where localised activity patterns propagate through superdiffusive Lévy flights. Such Lévy flight dynamics of the centre of mass of firing spikes $\mathbf{x}_t$ with a Langevin equation driven by Lévy noise and coupled to a momentum term $\mathbf{p}_t$, which we shall call the FNS equations of motion,

$$\begin{aligned} d\mathbf{x}_t &= \gamma b(\mathbf{x}_t)dt + \beta \mathbf{p}_t dt + \gamma^{1/\alpha} dL_t^\alpha, \\ d\mathbf{p}_t &= \beta b(\mathbf{x}_t)dt \end{aligned} \tag{1.9}$$

where $\mathbf{x}_t$ is the (usually 2 dimensional) vector of position, $\mathbf{p}_t$ is a generalised auxiliary variable analogous to a momentum of the same dimensionality, $\beta > 0$ is a parameter defining the strength of coupling between $\mathbf{p}_t$ and $\mathbf{x}_t$, $\gamma > 0$ is the noise strength parameter, $0 < \alpha \leq 2$ is the Lévy tail exponent, and dL_t^α is the Lévy stable noise drawn from a $\mathcal{S}\alpha\mathcal{S}$ distribution. The term $b(\mathbf{x}_t)$ captures a deterministic gradient term in the dynamics, defined such that the invariant distribution of $\mathbf{x}_t$ after long time t approaches the target distribution sampled, $\pi(\mathbf{x})$. The exact expression for $b(\mathbf{x})$ is derived in [27], and approaches the gradient of the logarithm of the density as $\alpha \to 2$, with the full form for any $0 < \alpha \leq 2$ given as

$$b(\mathbf{x}) = \frac{\mathcal{D}_{\mathbf{x}}^{\alpha-2}[\pi(\mathbf{x})\partial_{\mathbf{x}}\pi(\mathbf{x})]}{\pi(\mathbf{x})} \tag{1.10}$$

The FNS dynamics are called fractional, as the nonlocal dynamics of their distribution require fractional calculus to describe such spatial superdiffusion in Lévy flights [78]. The theory of FNS thus unifies the scale-free nonlocal dynamics of neural firing patterns with the neural sampling framework derived from the ideal Bayesian scheme of active inference. The critical advantage of FNS over other standard continuous random walks is its ability to efficiently sample complex, multimodal probability distributions with far-apart modes—likely a major feature of processed natural environments replete with multiple distinct salient features. While Brownian-motion-based samplers typically remain trapped in local modes due to high potential energy barriers, the

heavy-tailed, large jumps inherent in FNS enable the neural sampler to adaptively and freely switch between multiple modes [79–82]. Furthermore, the generality of the FNS description permits the application of FNS equations of motion to gaze dynamics, as the motion of the eyes are driven by an IE neural circuit which is expected to keep key features of the FNS model invariant, such as superdiffusion, heavy-tailed statistics, and nonlocal interactions.

1.2.2 Markov chain monte carlo methods

This view of neural- and visual sampling of a target distribution motivates analogy to a foundational set of methods in machine learning. Markov Chain Monte Carlo (MCMC) methods have become the algorithmic basis for generating samples from target distributions in rigorous Bayesian inference in machine learning [51, 80, 83, 84]. MCMC operates by constructing a Markov chain, where the transition matrix is calculated such that the equilibrium distribution of samples is the desired target distribution. In the context of neural computation, biological circuits are often modelled as implementing continuous-time variants of MCMC sampling of neural population targets, most notably through Langevin sampling and Hamiltonian Monte Carlo (HMC) [80, 85]. Langevin sampling dynamics utilise local gradient information from the probability landscape, combined with Gaussian white noise, to explore the state space over time. Such dynamics are identical to the Langevin system studied in statistical physics, described in Equation (1.1), where $V(\mathbf{x}) = -\log(\pi(\mathbf{x}))$ is the potential function over parameters $\mathbf{x}$ required yielding a distribution $\pi(\mathbf{x})$ of samples. Despite their ubiquity, traditional MCMC methods encounter severe theoretical and practical limitations when exploring unknown, heavy-tailed, and/or multimodal target distributions. Standard Langevin and HMC dynamics are driven by continuous Brownian and Hamiltonian motion respectively, and the traversal of low-probability regions, equivalent to dynamical energy barriers, becomes exponentially slow. Consequently, the mean exit time for the sampler to leave one mode and discover another grows exponentially with the modal separation distance [80]. To accelerate this sluggish exploration, algorithms often introduce auxiliary momentum variables to drive global landscape traversal, as in the FNS equations of motion 1.9. However, even with momentum-based acceleration, traversing deep energy valleys remains highly inefficient [79, 80, 85]. The nonlocal dynamics introduced in fractional sampling are one branch of research into how rapid mixing of samples in complex distributions can be achieved without the need for nontrivial parameter tuning and knowledge of the distribution of interest. For instance, fractional algorithms based on Hamiltonian Monte Carlo sampling and second-order Langevin sampling with a momentum variable have been briefly developed and explored [27, 81, 86]. Importantly, FNS dynamics over a probability landscape yield a mean exit time between modes which scales linearly rather than exponentially with separation [27]. FNS dynamics thus demonstrate a profound computational advantage for efficient biological inference through it's background as an effective sampling algorithm.

1.3 Behavioural data

To ground the physical background of this thesis, this section reviews the empirical behavioural data which characterise human visual inference. The visual system does not operate as a passive receiver of sensory input, but rather as an active, dynamic sampler of a changing environment [87]. This active sensing manifests across two coupled physical observables: the internal dynamics of rich neural oscillatory networks, and the external, mechanical execution of gaze trajectories. Crucially, these domains inform and drive each other, and each exhibit complex, scale-free dynamics in their activity. Characterised by background spectra with power law tails and anomalous diffusion, the visual system is capable of devoting attention to many spatiotemporal scales of event. Viewing these biological phenomena through the lens of statistical mechanics provides a rigorous foundation for mathematically modelling the visual system. This section establishes these empirical constraints by first examining the functional role of neural frequency bands and theta-gamma coupling (Section 1.3.1), before detailing the multi-scale, stochastic regimes of human gaze trajectories (Section 1.3.2). Together, these behavioural patterns moti-

vate the fractional dynamics given by the FNS equations of motion, and the developments upon that model in the subsequent sections.

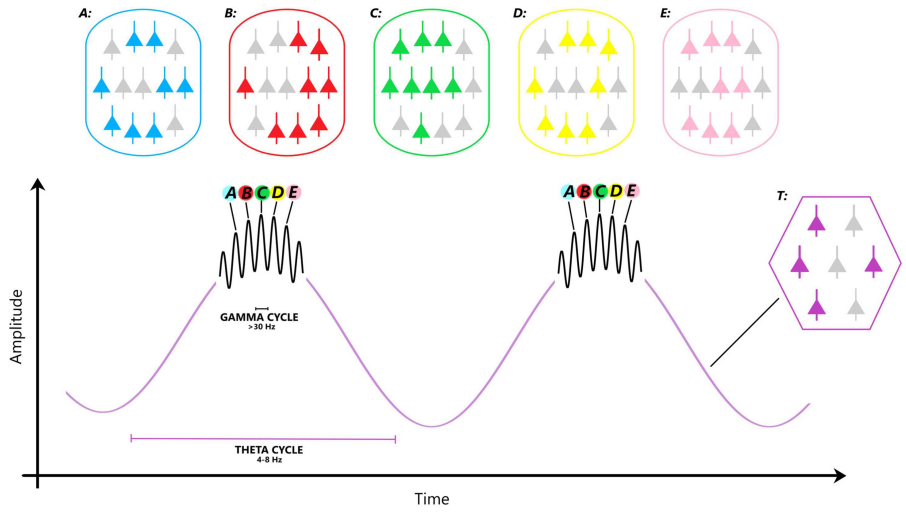


Figure 1.2: A schematic representation of theta-gamma coupling (TGC). Here, neural population patterns $\mathbf{A} - \mathbf{E}$ are activated in sequence on the timescale of a gamma oscillation $f > 30$Hz, triggered by the phase of a modulating theta wave of neuron population $\mathbf{T}$ on the theta timescale of $4 - 8$Hz. Such a formulation of TGC as correctly ordering signals has its foundations in the work of Lisman and Jensen [88], and was subsequently expanded to describe the dynamic neural mechanism during sensory exploration to modulate close attention upon individual centres of focus [89]. *Schematic adapted from figure 2 of Ursino and Pirazzini [90]*

1.3.1 Neural frequency bands

Throughout much of this thesis, links are drawn between timescales of interactions and dynamics in the FNS model and well-studied frequency bands of activity in the human brain. It is then valuable to outline the studied computational and functional interpretations of such frequency bands. Neural firing activity is divided into frequency bands named by greek letters, originally discovered and decomposed from electroencephalogram (EEG) data. Divisions are only approximate, but for this thesis these bands are: delta (0.1 to 4 Hz), theta (4 to 8 Hz), alpha (8 to 14 Hz), beta (14 to 30 Hz), and gamma (30 to 220 Hz) [91, 92]. Of particular interest to this thesis are the interpretations of theta and gamma neural activity as functionally important to attention and search tasks of computation. Theta rhythms have been shown to modulate attention during auditory tasks [93, 94], visual search tasks [95–97], and in analysing the general attentional mechanism of the brain [91, 92, 98]. Intervals between theta peaks are finding stronger interpretation in the neuroscience literature as discretely sampling the environment for integration into the internal model of the brain. Strong direct evidence for this theory include correlations between intelligence and high theta peak power in EEG data during tasks requiring focus and attention [93, 98], and the discovered sensitivity of the visual system to react and search scenes on a theta rhythm [95, 96]. Accordingly, the firing patterns which define the lowest level of information carried in the brain are regularly found to be gamma in scale, almost always within 30 to 100 Hz [90, 99–102]. The coupling between the phase of theta activity and the amplitude of gamma activity is known as theta-gamma coupling (TGC). This phenomenon, a specific form of phase-amplitude coupling (PAC), is schematically represented in Figure 1.2. The evidence and theory behind TGC are described in a systematic review by Ursino and Pirazzini [90]. This thesis provides a computation view of the origin and function of TGC in the visual system as modulating periods of attention during which fixations exhibit dynamics of active sensing.

1.3.2 Gaze trajectories

Healthy human gaze trajectories consist of different dynamical regimes. The most studied pattern of gaze trajectories are their division into periods of fixation, interspersed by near-instantaneous jumps between fixations, these jumps being known as *saccades* [103–106]. Importantly, this behaviour of local movements interspersed with large discontinuous jumps suggest modelling eye movements through a Lévy flight. This paradigm has a limited history in the literature, and fitting data to a simple phenomenological Lévy flight parameterised by the stable parameter α has found some limited use as a biomarker for ADHD [107]. Broader saccade metrics have been found to be valuable as non-invasive biomarkers for broader neural disorders, including schizophrenia [108–111], autism [112, 113], OCD [112], bipolar disorder [112, 113], and Parkinson's disease [112, 113]. The dynamics of fixation have received less scholarly attention as a biomarker of cognition. Gaze fixations exhibit three distinct dynamical patterns: near-instant microsaccades within the same point of focus, slow ocular drift about a point of focus, and high frequency ocular micro tremor [114–117]. Detailed examinations and systematic reviews of these behaviours have not considered their computational role in active sensing and efficient search. The work presented in this research strongly motivate a closer and more biologically informed understanding of the dynamics of gaze trajectories, across a number of spatial and temporal scales, to diagnose and understand the function of global functioning of the brain, much of which controls the visual system.

Chapter 2: Methods

2.1 First section

2.1.1 First subsection

2.2 Second subsection

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